



Reappraising the roles of review valence and conflict in online relationships

Ran Liu^{a,*}, John B. Ford^b, Weiyong Zhang^c, Joseph Bonnici^d

^a Central Connecticut State University, 465 Vance Academic Center, New Britain, CT 06050, USA

^b Old Dominion University, 2117 Constant Hall, Norfolk, VA 23529, USA

^c Old Dominion University, 2072 Constant Hall, Norfolk, VA 23529, USA

^d Central Connecticut State University, 461 Vance Academic Center, New Britain, CT 06050, USA

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ABSTRACT

The present work addresses the influence of review valence on consumers' perceived relationship with a firm and identifies the causes of such influence and the underlying process. Based on social exchange theory and signal theory, the authors identify how conflict is pivotal in connecting review valence and consumers' perceived relational benefits and trust. The authors use Qualtrics survey data in three experimental studies to examine how review valence indirectly impacts consumers' perceived trust and benefits. The results indicate that conflict is a moderator between review valence and perceived benefits and trust, and the magnitude of the indirect effect of valence varies according to the different levels of a perceived partner's commitment to the relationship. These findings provide incremental theoretical implications for relationship marketing in online contexts and have managerial relevance for online marketing strategies.

1. Introduction

The present work examines an invisible factor overlooked by previous research: How a negative online review may undercut perceived trust in a brand, a cornerstone of the consumer-firm relationship (R. Liu et al., 2022). Furthermore, the valence of brand-related electronic word-of-mouth (eWOM) practices affects short-term marketing outcomes, including purchasing behavior and referral intentions (Baber et al., 2016; Lim et al., 2022).

While eWOM's direct impact on marketing outcomes has been well documented, its indirect effects remain ambiguous (Hennig-Thurau et al., 2010; R. Liu et al., 2022; Palmatier et al., 2006). One reason for the neglect of the indirect impact is a lack of theoretical justification for the direct linkage between a short-term behavior and a long-term-oriented relationship outcome (Ganesan, 1994). For instance, relationship marketing (RM) outcomes (e.g., customer loyalty, benefits, trust) might not directly result from WOM activities. In other words, merely being exposed to positive or negative eWOM might not affect customer loyalty, which is a long-term concept (V. Kumar, 2010).

Driven by social exchange theory (SET) (Blau, 1964) and signal theory (Kelley, 1988), this study challenged such assumptions and

investigated how eWOM may serve as an antecedent to RM. First, we identified perceived conflict as a mediator connecting eWOM valence and one fundamental RM construct: relational benefits and trust, establishing a significant and indirect link between a short-term activity and a long-term perceptual attitude (Hennig-Thurau et al., 2002). Based on the findings of experimental studies covering three major service industries (banking, airlines, and hotels), we explored the underlying processes of how being merely exposed to negative information impacts an individual's perception of a relationship (Study 1), how conflict mediates the linkage between valence and relational benefits and trust (Study 2), and the boundary conditions under which the indirect effect would be significant (Study 3).

Our results make incremental contributions to the previous literature. First, eWOM not only affects consumers' short-term decisions (Anderson & Salisbury, 2003) but also alters their perceptions regarding potentially long-lasting customer-firm relationships. Thus, eWOM valence serves as an antecedent to RM, not only an RM outcome (Palmatier et al., 2006). Second, we found that perceived conflict—the single antecedent construct that negatively impacts all relational factors—mediates eWOM valence and benefits and trust, thus contributing to the RM literature by identifying new approaches to mitigating

Abbreviations: B2B, business-to-business; B2C, business-to-consumer; CI, confidence interval; CFA, confirmatory factor analysis; eWOM, electronic word of-mouth; NWOM, negative word-of-mouth; OLS, ordinary least squares; PWOM, positive word-of-mouth; RM, relationship marketing; SET, social exchange theory; UGC, user-generated content.

* Corresponding author.

E-mail addresses: liu@ccsu.edu (R. Liu), jbford@odu.edu (J.B. Ford), wyzhang@odu.edu (W. Zhang), bonnicij@ccsu.edu (J. Bonnici).

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conflict. Third, the work contributes to SET and signal theory by applying traditional relationship constructs to the online context. This confirms the importance of online information valence in balancing a reciprocal relationship (Kozlenkova et al., 2017; Liu et al., 2021).

Finally, we discuss how our findings offer meaningful implications for how online relationships may be adversely affected by mere exposure to a negative review. The boundary conditions of the valence effect highlight the importance of using different marketing strategies for varied marketing segments. While our findings are based on online review valences from rating websites, external validity can be extended to all online media platforms, demonstrating the growing focus on building virtual relationships (Steinhoff et al., 2018).

2. Theoretical background and hypotheses development

eWOM serves as the most common communication platform for consumers to interact with firms (King et al., 2014; Yadav & Pavlou, 2014) and is one of the most potent influencers of consumers across all stages of the decision-making process, including awareness, interest, and the final decision (De Bruyn & Lilien, 2008; Moon et al., 2010). Moreover, among all types of user-generated content (UGC), eWOM has a significant impact on online shopping behavior (Pavlou & Dimoka, 2006) and firm performance, including e-commerce success (Kozlenkova et al., 2017; Trusov et al., 2009).

The eWOM matrix—such as valence, volume, variance, the strength of ties, source expertise, and perceived affinity—affects the extent to which eWOM influences consumers (Babić Rosario et al., 2016; Bansal & Voyer, 2000; Berger & Milkman, 2012). Among all factors, valence and volume are the most salient determinants of eWOM's effects (Babić Rosario et al., 2016). eWOM valence, which refers to the direction of the information, is often expressed positively (positive WOM, or PWOM), negatively (negative WOM, or NWOM), or neutrally. Negative information is an essential cue in eWOM communication and is perceived to be more informative and diagnostic than positive information (Lynch, 2006; Tang, 2010). While several researchers have demonstrated this negativity effect, this phenomenon is far from conclusive, and there are contradicting views in the previous literature (Chintagunta et al., 2010). For example, Hennig-Thurau et al. (2015) found that negative information is more powerful than positive information. Ahluwalia et al. (2000) found that negative information for highly committed consumers with a brand is less diagnostic, and therefore they are less affected by it. In addition, Fitzsimons and Lehmann (2004) found that consumers became more likely to choose a product when they read negative information about it if it contradicted their initial preference. More studies have confirmed the existence of the negativity effect and that as valence becomes more extreme, information becomes more helpful in reducing consumers' attitudinal uncertainty (Chua & Banerjee, 2016). The

ambiguity of prior findings on eWOM valence provides a rationale for reappraising the interaction between brand reputation and review valences (Wilson et al., 2017).

2.1. RM in online relationships and manifesting conflict

RM connects system-oriented marketing concepts such as commitment, trust, conflict, and benefits, which are critical in cultivating company-customer ties and determining marketing outcomes (Grönroos, 1995; King et al., 2014; see Fig. 1). Traditional literature on RM underlines the importance of the business-to-business (B2B) relationship as a research context because the channel partners are more likely to make a business decision based on the level of interdependence and coordinated interaction among the players (Anderson & Weitz, 1989). The rising popularity of online interactions (Internet-based relationships) has become the dominant exchange mode for most consumers. It has led to new forms of company-customer bonds (Steinhoff et al., 2018). These online relationships provide both opportunities and challenges for RM managers. On the one hand, web-oriented customer segmentation has increased sales volume and favors more customer communication channels (Ansari et al., 2008). On the other hand, lower search costs and online anonymity create concerns for marketers (Shankar et al., 2003). As e-commerce has become a predominant focus of online exchange, consumers seek online information to evaluate the credibility of their business partners before making any transactions (Kim & Hanssens, 2017). For example, in the hospitality sector, 148.3 million travel bookings are made on booking websites, and 80% of all users will read other travelers' reviews (Condor Ferries, 2020). Approximately 94% of travelers claim that reviews are critical when choosing hotels, especially when checking a service provider's reputation (TripAdvisor, 2020). Nevertheless, the direction of causality of the relationship between eWOM and RM has yet to be verified (Hennig-Thurau et al., 2010; Palmatier et al., 2006). Mutual communication between firms and consumers is an antecedent to RM, and WOM is a consequence of RM efforts (Hennig-Thurau et al., 2002; Morgan & Hunt, 1994). In the online context, the richness of communication channels (e.g., online reviews, social media discussions, mobile apps) provides new characteristics of communication, which has transformed previously two-way interactions into omnichannel using a multi-partner approach (Gelb & Sundaram, 2002; Verhoef et al., 2015). In addition to containing richness, eWOM interactions differ from traditional offline communication in ways such as confidentiality, freedom from geographic bias, fewer time constraints, and the permanence of conversation (Gelb & Sundaram, 2002). Despite these technological changes, the WOM-as-consequence RM model remains a dominant perspective, and there is scarce research on how WOM affects RM factors (De Matos & Rossi, 2008; Palmatier et al., 2006). There is a research gap in

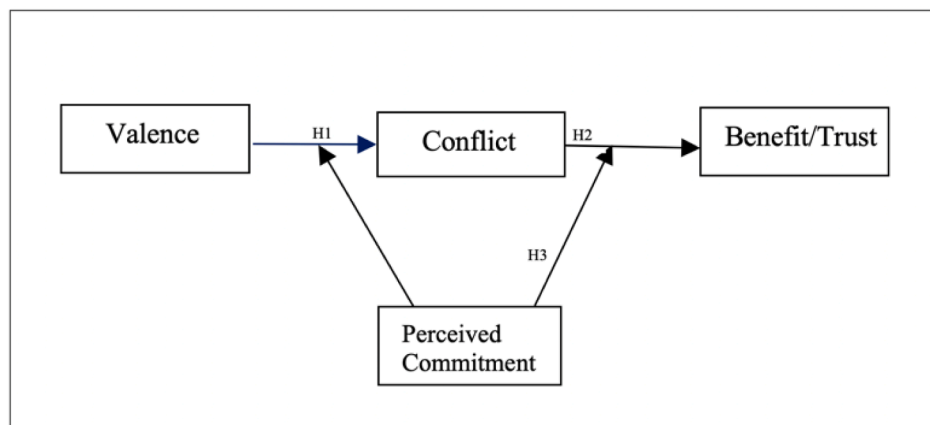


Fig. 1. Conceptual Model.

identifying potential antecedent roles through online communication (including eWOM) in establishing and enhancing online relationships and finding the critical constructs in determining this process (Hennig-Thurau et al., 2010; Yadav & Pavlou, 2014).

The impact of conflict on relationship marketing (RM) is stronger and more negative than any other relational construct (N. Kumar et al., 1995; Palmatier et al., 2006). The presence of conflict denotes a decreased level of distributive fairness (fairness of results), procedural fairness (fairness of process), or both and, therefore, undercuts relationship quality (N. Kumar et al., 1995). Conflict is viewed as a multi-stage process composed of different phases of disagreement with partners (Brown et al., 1991), such as latent, perceived, affective, and manifest conflict, as well as conflict aftermath (N. Kumar et al., 1995). Latent conflict refers to the inherent distrust and tension between relational partners; perceived conflict is the awareness of these tensions; affective conflict is when this awareness is transformed into hostile feelings; and manifest conflict refers to the stage where the existence of disagreements is confirmed and can be measured by the intensity of overall disagreement with a partner. These processes result in conflict aftermath, which becomes a conflict episode for future latent disputes (Brown et al., 1991; Duarte & Davies, 2003). These conflict stages often overlap with each other and are not necessarily sequential. For example, a latent conflict is not necessary to form a perceived conflict, and a perceived conflict may be resolved before any manifest conflict happens (Duarte & Davies, 2003; Mohr & Spekman, 1994). Manifest conflict is one of the major concerns for RM marketers and has been identified as an essential dyadic antecedent; it plays a crucial role in determining relationship quality and firm performance (N. Kumar et al., 1995; Palmatier et al., 2006). Moreover, consumers have a psychological self-defense system to protect their self-integrity from outside information and an inherent desire to pursue fairness among exchange partners. Thus, examining the association between eWOM and conflict becomes critical in understanding the potential new role of eWOM in online RM (Grayson, 2007). Given the high relevance to marketing (N. Kumar et al., 1995), manifest conflict is used as the proxy of conflict in online relationships.

As a fundamental organizational and relationship exchange theory, SET plays a significant role in connecting multiple disciplines, such as social psychology (Gouldner, 1960), sociology (Blau, 1964), and anthropology (Sahlins, 1972). SET helps to explain the psychological mechanism connecting eWOM and online RM constructs since it involves how elements of interdependent social transactions potentially affect relationship quality and how they are influenced by other people's behavior (Blau, 1964; Cropanzano & Mitchell, 2005; Meeker, 1971). SET argues that when individuals exchange resources (e.g., goods, services, money, information, status, love), the exchange process depends on a preserved justification for all exchange partners (Foa & Foa, 1980; Simonson, 1989). In a fair and reciprocal relationship, consumers receive goods or services that meet or exceed their expectations (as in PWOM) so that the exchange process maintains or enhances the relationship (Cropanzano & Mitchell, 2005). In contrast, the perception that the exchange process lacks fairness and reciprocity might jeopardize an existing relationship, thus leading to increased conflict, as often seen in NWOM (Graen & Uhl-Bien, 1995). This argument also aligns with signal theory, which explains that signals sent by consumers or sellers in online shopping communities play vital roles in helping buyers identify credible sellers and form bilateral relationships (Kozlenkova et al., 2017). NWOM indicates to potential buyers that the exchange partners do not treat each other well and that there might be conflict in the relationship (Blodgett et al., 2001; Kozlenkova et al., 2017).

PWOM is motivated by altruism, product knowledge, self-enhancement, and a reciprocal relationship with the company, while NWOM is often propelled by anxiety, a negative experience, and the need for advice from others about one's unfortunate situation (Sundaram et al., 1998). As a result, NWOM signals that there are potentially unfair situations or increased risks with social transactions.

Nevertheless, the impact of risk-increasing signals varies and is contingent on the consumer's psychological commitment to the relationship. SET shows that the exchange partners' level of commitment is one of the most promising factors distinguishing social exchanges from economic ones (Bishop et al., 2000). Commitment results from reciprocal efforts in social exchange; however, the causal direction is reversible.

Wiener (1982) treated commitment as a motivational factor bridging behavioral antecedents and consequences. N. Kumar et al. (1994) provided a twofold explanation of commitment: affective and calculative commitment. They claimed that affective commitment measures the level of commitment based on an exchange partner's desire to keep a relationship and an overall positive feeling. Calculative commitment refers to an exchange partner maintaining a relationship based on calculating its risks and benefits. This is usually treated as a negatively oriented motivation because the relationship is maintained due to the lack of availability of an alternative relationship (Allen & Meyer, 1990; N. Kumar et al., 1994). Gruen et al. (2000) divided calculative commitment into continuance and normative commitment. They stated that although both are rooted in perceptions of cost and benefits, continuance commitment focuses on the psychological linkage to a relationship partner, while normative commitment centers on the moral obligation to maintain a relationship. East et al. (2008) found a close connection between the impact of WOM and the prior exchange probability. They discovered that consumers who are more committed to making a purchase are unlikely to be affected by NWOM, and consumers who are less committed to buying something are less affected by PWOM. Their findings illustrate that consumers respond to information differently based on their prior willingness to establish or maintain a relationship. Compared with consumers not interested in maintaining the relationship (low level of commitment), those who intend to retain it (high level of commitment) would be more concerned regarding the negative issues raised by NWOM. Nevertheless, it is unclear how the perception of a company's commitment would affect the relationship balance. Because Perceived partners' commitment is correlated but different than brand commitment, it is a separate construct for measuring relationships. Specifically, perceived partners' commitment indicates the perception of a partner's desire to maintain and improve confidence in the stability of a relationship (Jap and Ganesan 2000) rather than the self-motivation to maintain the relationship (relationship commitment). Thus, we argue that there is a joint effect of perceived commitment and review valence on conflict (see Fig. 1):

H1: The impact of review valence on conflict is moderated by consumers' perception of their partner's level of commitment to the relationship, such that the effect of review valence is stronger when consumers perceive that their partner has a high level of commitment.

2.2. The mediating role of conflict

A stable relationship among social exchange partners requires members to share similar perceptions and balanced fairness, reflecting consumers' judgments of a firm's behavior and services (Blau, 1964; Narteh, 2016). When consumers are exposed to negative online information, according to SET, their balanced perception(s) will be altered, and a cognitive re-evaluation of the exchange relationship will occur. In online contexts, a conflict caused by negative information can be more severe due to the asymmetrical information and limited resolution techniques provided (Mohr & Spekman, 1994). For example, an emerging conflict can be resolved offline through face-to-face communication and real-time information sharing. However, this type of joint problem-solving is challenging to recreate in online contexts (Mohr & Spekman, 1994). Consequently, the increased level of conflict induced by negative online information will automatically alter the consumer's perception of the partner, undercut their confidence in the partner, and reduce the level of trust (Hennig-Thurau et al., 2002; N. Kumar et al., 1995; R. Liu et al., 2021). In other words, SET maintains that mutual

understanding and positive emotions are part of social exchange resources (Foa & Foa, 1980). Thus, intensified conflict undermines these resources, leading to a diminished level of perceived relational benefits. Collectively, this discussion led us to develop the following hypothesis:

H2: The effect of review valence on relational benefit and trust is mediated by the intensity of the conflict consumers feel

2.3. The moderating role of the perceived partner's commitment

Perceived conflict (or simply conflict) represents consumers' overall disagreement with exchange partners and serves as a critical dyadic antecedent of RM (N. Kumar et al., 1995). Conflict is an essential factor that negatively affects customer relationships among all dyadic relational antecedents. As consumers may feel they are under attack when they are exposed to negative online WOM because their loyalty to the brand and their positive self-view are challenged (Cheng et al., 2012), the presence of conflict significantly undermines the positive effects of other relational antecedents (Anderson & Weitz, 1989; Palmatier et al., 2006). These findings are congruent with Gilbert et al.'s (1998) study, suggesting that people have psychological immune systems to defend their self-integrity from environmental threats. Most importantly, increased conflict can lead to a loss of confidence in an exchange partner and decrease business interest (Grayson, 2007; Morgan & Hunt, 1994), especially for tangible and high-financial-risk products on e-commerce platforms (Babić Rosario et al., 2016). According to SET, increased conflict is negatively associated with trust along different dimensions, including a firm's ability, benevolence, and integrity (R. Liu et al., 2022). This may be because resource exchange involves economic and socioemotional outcomes. In other words, as SET indicates, a tangible financial reward is not the only factor that matters; exchange partners also seek respectfulness, fairness, and trust in the relationship (Foa & Foa, 1980; R. Liu et al., 2022).

Trust involves the exchange partner's reliability and integrity and is associated with the fiduciary obligation to prioritize a relational partner's interests (N. Kumar et al., 1995). As perceived conflict increases, the balance of reciprocity between exchange partners deteriorates, causing partners to be viewed as less respectable or trustworthy (Verma et al., 2016). In online contexts, relationships' spatial and temporal distances lead to information asymmetries; this underlines the importance of trust in forming a reliable relationship (Pai & Tsai, 2011). Consumers are willing to pay for a transaction based on the expectation that it will be rewarding and the trust that the exchange partner will make the relationship fair and reciprocal (Cropanzano & Mitchell, 2005). However, as SET and signal theory indicate, NWOM sends the opposite unfavorable implication and signals that individuals are not being treated fairly by their exchange partners; this unfairness will lead to a decreased level of trust (Colquitt et al., 2001).

At the same time, consumers use trust as a decision-making tool to minimize risks and maximize benefits (Lohmann, 2000). SET suggests that perceived benefits motivate consumers to strengthen a relationship and that only a reciprocal relationship where exchanges benefit both partners can be maintained (Hibbard et al., 2001). In an online relationship, as the consumer's conflict increases, the relationship becomes less reliable; hence, as Fig. 1 demonstrates, there is a main adverse effect of conflict on perceived benefits.

Although perceived conflicts undercut consumers' perceived benefits and trust, the magnitude of the effect is different depending on how strongly consumers believe their partner wants to preserve the relationship, which can be reflected in commitment (Verma et al., 2016). Commitment is a relational construct that substantially affects RM (De Matos & Rossi, 2008). SET denotes that social exchange alters commitment, which also determines the relationship between exchange partners (Bishop et al., 2000). Since commitment reflects a relational partner's willingness to stay (affective commitment) and their judgment of potential risks and benefits (calculative commitment) (N. Kumar et al., 1994), the perceived conflict induced by negative information

would affect individuals differently depending on their perception of the partner's commitment to the relationship, which indicates the perception of a partner's desire to maintain and improve confidence in the stability of a relationship (Jap and Ganesan 2000). While perceived partners' commitment is conceptually different from traditional relational commitment per se, the perception of the other party's commitment is highly related to the actual level of commitment from each party (Anderson & Weitz, 1992). For brevity, we used the term "perceived commitment" in this work with the recognition that it indicates the perceived partners' commitment. In other words, the perceived commitment in this work indicates the consumers' perception of the service provider's commitment to the relationship.

Perceived commitment is positively related to consumers' confidence in the relationship outcomes. Increased perceived commitment implies that consumers believe firms take the relationship seriously and are willing to maintain and improve it (Jap and Ganesan, 2000). Therefore, it is essential to examine how perceived commitment is involved in the process when conflict undercuts relational benefit/trust, which is one of the foundations for relational stability (Chung & Shin, 2010). In line with the previous discussion, conflict occurs when consumers believe that their interests or intentions are being disagreed with or even rejected by a company. Nevertheless, when consumers believe the firm has a high commitment level, they are less likely to doubt its behavior even in a conflicting situation and more likely to judge the firm benevolently (Boon & Holmes, 1991; Morgan & Hunt, 1994). As a result, conflicts are often handled with a functional solution when the perceived commitment is high. Thus, the adverse effects of the conflict are often offset by the benevolent perception of the firm. For example, if a customer had a conflict with hotel staff due to a long wait to check in, this may cause increased distrust about the quality of the hotel's service. However, if the hotel showed their seriousness about the mishandling of the situation and made efforts to mitigate the damage, i.e., providing price discounts or free room upgrades, the customer would believe that the hotel is committed to their guests and the distrust resulting from the delay would be mitigated compared to if the hotel did not take any action. As the example demonstrates, while increased conflict undermines consumers' confidence in the relationship's stability, a higher level of perceived commitment would attenuate the detrimental effects of conflicts on relationship outcomes. As such, we formulated the following hypothesis:

H3: The impact of conflict on relational benefits and trust is moderated by consumers' perception of the firm's commitment, such that the negative effect of conflict is stronger for consumers with strong beliefs that the firm is committed to the relationship than those with weak beliefs.

3. Method and results

We conducted three experiments in a laboratory setting to study the impact of review valence on online business-to-consumer (B2C) relationships across three service industries: banking, airlines, and hotels. These three sectors represent different service categories and have different levels of relationship termination costs, which are one of the most significant bonding forces, per the RM literature (Bowen, 1990; Sharma & Patterson, 2000). Specifically, financial services have high termination costs and an elevated level of information asymmetry; therefore, the bank-customer relationship is mainly dependent on trust, which is one of the focal constructs of the present work (Sharma & Patterson, 2000). In addition, the restaurant and hotel industries have been used in prior research as valid proxies for testing the impact of online reviews due to the increasing popularity of review websites such as Yelp and TripAdvisor (Grewal & Stephen, 2019; Mayzlin et al., 2014). Among the three studies, Study 1 examined how exposure to reviews with different valences could affect consumers' psychological conflict toward the relationship with their bank. Study 2 focused on the effect of review valence on relational benefits and trust with airline services.

Finally, Study 3 gauged hotel services, where conflict was introduced as a mediator for the impact of review valence on perceived relational benefits and trust.

3.1. Experimental design

To identify the effects of review valence on focal relational constructs and their boundary conditions, we conducted a lab experiment with the manipulation of the review valence (positive vs. negative). Using manipulations in lab experiments has been well-documented as a practical approach to testing the impact of review valence (Berger, 2011; Grewal & Stephen, 2019). To identify the effects of review valence on online relational constructs, we used Qualtrics to collect survey data. We used this online sampling platform because it provides quality control algorithms, such as meta-data-based relevant ID and time tracking, to prevent fraud in data screening (Arndt et al., 2022).

Using a between-subjects design, we randomly assigned participants to one of two conditions (positive or negative reviews). Participants were informed that the study's purpose was to understand how online reviews could affect their relationship with their primary bank (airline and hotel in the subsequent studies). They were asked to identify which bank they were using and to imagine that the reviews they would read were about their bank. Before they examined the reviews, they were asked to indicate their perception of their bank's level of commitment to the relationship with them (perceived commitment) and their emotional status using a 7-point scale (1 = "extremely negative," 7 = "extremely positive"). After reading the reviews, the participants' intensity of conflict with their banks was assessed before they provided demographic information (e.g., age, gender, income, and education level).

We used all measures—perceived valence, conflict, commitment, and relational benefits and trust—in this study and adapted all measures from previous literature. We performed factor analysis to check loadings and validities via a separate pre-test (see Table 1). We used the 7-point scale that Berger (2011) developed to check valence manipulation. We assessed the participants' perceived valence regarding the online reviews by asking them to rate the reviewers' feelings when they wrote those reviews using a 7-point scale, with −3 indicating "extremely negative" and 3 implying "extremely positive" (Berger, 2011).

Table 1
Scale Summary.

Constructs and Measures	Factor Loading
Conflict	AVE = 0.87 α = 0.92
1. My relationship with the bank can be best described as tense.	0.88
2. I have significant disagreements in my relationship with the bank.	0.86
3. I frequently clash with the bank on issues relating to how they should conduct their services.	0.86
Perceived Partner's Commitment	AVE = 0.65 α = 0.75
1. The bank is quite willing to dedicate whatever people and resources it takes to help me.	0.57
3. The bank is continually on the lookout for other customers and ignores existing ones.	0.83
4. It takes me too much time, effort, and energy to get the bank to resolve my problems.	0.92
5. The bank's services are not as expected compared to other bank(s).	0.86
Confidence in Benefits and Trust	AVE = 0.72 α = 0.87
1. I know what to expect when I use this bank.	0.75
2. This bank's employees are perfectly honest and truthful.	0.87
3. This bank can be trusted completely.	0.91
4. This bank has high integrity.	0.85

Notes: α : Cronbach's α ; AVE: average variance extracted; CFA: confirmatory factor analysis.

For Studies 2 and 3, the word "bank" was changed to "airline" or "hotel," respectively.

We assessed the intensity of the conflict using the scale proposed by Jap and Ganesan (2000), who employed it to gauge retailers' perceptions of the conflict level with suppliers. Given the sound reliability documented (Cronbach's alpha: 0.83) and the meaning of the three items, we found the scale suitable to measure the level of conflict in the bank-client context. Specifically, participants reported their level of agreement with the three items on a 7-point scale (1 = "strongly agree," 7 = "strongly disagree"). Likewise, the level of the partner's commitment.

was assessed using Jap and Ganesan's (2000) scale. Participants indicated their level of agreement with four items on a 7-point scale ($\alpha = 0.75$). In addition, as a proxy for measuring benefits and trust, we used the 4-item scale developed by Hennig-Thurau et al. (2002) to measure service industry relationships (see Table 1).

3.2. Pre-test and manipulation check

We conducted a pre-test with factor analysis and examined the construct reliability and validity based on data from 101 participants gathered through Amazon Mechanical Turk (out of an initial 110) in the US. We used the average variance extracted (AVE) and Cronbach's alpha as proxies to test the scales' validity and reliability (Kline, 1998). Previous literature suggests that an AVE with a value > 0.50 supports convergent validity (Bagozzi & Yi, 1988). Based on this rule, the test results suggest that both the scales used in the study are valid (AVE > 0.50) and reliable ($\alpha > 0.70$). The items had standardized loadings ranging from 0.57 to 0.92, which are within the acceptable level (Kline, 1998). The results of the factor analysis are outlined in Table 1. In addition, we confirmed that the construct correlations are significantly smaller than each construct AVE, providing support for discriminant validity (see Tables 1 and 2). Finally, a *t*-test revealed no significant difference in the level of commitment before and after participants read the manipulated reviews, meaning that the level of commitment was not affected by review valence.

After assessing the measures, we performed a one-way between-subjects analysis of variance on perceived valence as a function of valence (positive vs. negative). The results imply that the effect of valence was significant ($F(1,80) = 109.55, p < 0.001$), and participants in the positive condition believed that the manipulated reviews were more positive. In contrast, those in the negative condition believed the reviews were more negative ($M_{\text{positive}} = 5.72, M_{\text{negative}} = 2.67$).

3.3. Study 1

3.3.1. Participants and procedure

After checking the manipulation and measurement validity, we recruited 200 Qualtrics members as participants for the online survey and gave them a nominal payment ($M_{\text{age}} = 38.58$ years, 36% female). Then, we randomly assigned all participants to two valence conditions; they answered questions regarding the dependent variables. Specifically, 166 participants passed the attention checks (i.e., asking participants to choose a particular answer, such as "strongly disagree" after asking them a question); we classified their responses as valid.

Since banking settings have been used in previous RM literature for the service industry (Taleghani et al., 2011), we manipulated negative and positive review conditions based on the star ratings of bank reviews. In the positive condition, there were two manipulated 5-star reviews, and the other had a decoy 4-star review. In the negative condition, there were two manipulated 1-star reviews and a 2-star decoy review (see Appendix A). Those were actual bankrate.com reviews; we selected them because they were emotionally equivalent and had average length. No reviewer information was provided, and three reviews were under each condition. We created decoy reviews with moderate valence to mimic a real online rating scenario and moderate the extremity.

3.3.2. Results

We predicted that the negative (positive) reviews would increase

Table 2
Descriptive Statistics and Correlations of Key Constructs.

	Mean	SD	1	2	3	4	5	6	7
1. Perceived Conflict	4.14	1.31	1						
2. Perceived Commitment	4.54	1.37	−0.38**	1					
3. Benefits and Trust	5.01	1.10	−0.09	0.03	1				
4. Age	39.00	10.50	−0.27**	0.23**	−0.10	1			
5. Gender	n/a	n/a	−0.02	−0.10	−0.05	−0.17*	1		
6. Education	4.62	1.14	0.17*	−0.05	−0.07	0.01	0.01	1	
7. Emotional Status	5.05	1.49	0.06	0.22**	0.25**	0.10	0.10	0.05	1

* $p < 0.05$, ** $p < 0.01$ (two-tailed), $N = 101$.

(decrease) conflict only when participants believed that their bank had a high level of commitment. To test this, we regressed conflict on review valence (negative = 1, positive = 0), level of commitment, interaction, and control variables (age, gender, education level, and emotional status). Then, we ran an ordinary least squares (OLS) regression (see Model 1 in Appendix B) with conflict as the dependent variable; review valence (positive vs. negative), level of commitment, the interaction of valence, and commitment were the independent variables, and a set of variables comprised the control variables.

We used a three-model approach to report the moderating results (Bstieler et al., 2017). In model 1, the control variables were regressed on conflict. The results show that Model 1 was insignificant ($R^2 = 0.05$, $F(4,162) = 1.92$, $p = 0.11$), indicating that more variables and alternative modeling are needed to explain the variation in conflict. The only control variable that impacts conflict was education, which was positively associated with conflict ($b = 0.2$, $t = 2.38$, $p < 0.05$). Finally, the main effects were tested in Model 2, in which the model was significant ($R^2 = 0.29$, $F(6,160) = 12.59$, $p < 0.001$), and valence and commitment were significantly associated with conflict. While negative valence increased conflict ($b = 0.51$, $t = 7.73$, $p < 0.001$), a higher perceived commitment was negatively related to conflict ($b = -0.19$, $t = -2.73$, $p < 0.05$).

Model 3 added the moderating effects of perceived commitment on the association between review valence and conflict. The overall model was significant ($F(7,159) = 12.79$, $p < 0.001$), and there was no main effect of valence ($b = -0.09$, $t = -0.42$, $p = 0.68$). However, the interaction effect of valence and perceived commitment was positive and significant ($b = 0.65$, $t = 2.80$, $p = 0.006$, see Table 3). Notably, the effect of review valence was stronger when perceived commitment was high, thus supporting Hypothesis 1.

The findings of this study illustrate that the perception of a relational partner's commitment plays a critical role in determining the magnitude of the impact of review valence on conflict. Negative reviews increase consumers' perceived conflict and threaten to deteriorate the relationship only when they believe their partner is highly committed to the relationship. Thus, in the following study, we examined how perceived benefits and trust could be damaged due to negative reviews and the

boundary conditions of those impacts.

3.4. Study 2

Study 1 clarifies how online review valence has a causal impact on a critical relational mechanism: perceived conflict. Study 2 aims to deepen these insights by scrutinizing the mediation role of conflict between review valence and relational benefits and trust.

3.4.1. Design and procedures

To test H2, we adopted a similar procedure and experimental design with another service context: the airline industry. We recruited another 190 Qualtrics members and gave them a nominal payment; 155 participants passed the attention checks ($M_{age} = 37$ years, 42% female).

Participants were asked to identify an airline they were familiar with and report their perception of its commitment to them before they looked at the reviews. The stimuli used in Study 2 were developed using a similar approach to Study 1 (see Appendix C). To enhance realism, we drew the language used in these reviews directly from airlinequality.com, and half the reviews had a label displaying the number of reviews under each condition. Before they read the reviews, we asked the participants to indicate their emotional status using a 7-point scale.

After this, we asked the participants to read the reviews and imagine they were planning to choose an airline for travel. Participants then answered questions about perceived conflict (the same items from Study 1; $\alpha = 0.92$) and their perceived trust in, and benefits of, the airline (the same items from Study 1 with an altered context, $\alpha = 0.87$). Finally, the participants answered attention checks and standard demographic questions in random order.

3.4.2. Results and discussion

To determine whether conflict mediates the relationship between review valence and benefits and trust, we used PROCESS Model 8 (Hayes, 2018) with 5,000 bootstrap samples to estimate a moderated mediation model (see Fig. 2).

We observed a significant index of moderated mediation ($b = -$

Table 3
Multilevel regression analysis on Conflict.

	Model 1		Model 2		Model 3	
	Estimate	p-Value	Estimate	p-Value	Estimate	p-Value
Age	−0.07(0.01)	0.37	−0.08 (0.01)	0.28	−0.06 (0.01)	0.39
Gender (Female)	−0.05(0.28)	0.51	−0.03 (0.23)	0.63	0.01 (0.23)	0.90
Education	0.20(0.09) *	0.02	0.21** (0.08)	0.003	0.22** (0.08)	0.002
Emotion status	0.02(0.08)	0.80	0.03 (0.08)	0.66	−0.05 (0.08)	0.68
Valence (Negative)			0.51*** (0.46)	<0.001	−0.09 (−0.72)	0.73
Commitment			−0.19** (0.08)	0.007	−0.38*** (−0.12)	<0.001
Valence (Negative) * Commitment					0.65** (−0.16)	0.006
Adjusted R-square	0.05		0.29		0.32	
F-statistic	1.92(4,162)	0.11	12.35 (6,160) ***	<0.001	12.37 (7,159) ***	<0.001

* $p < 0.10$. ** $p < 0.05$ *** $p < 0.001$.

Notes: Standard errors in parentheses.

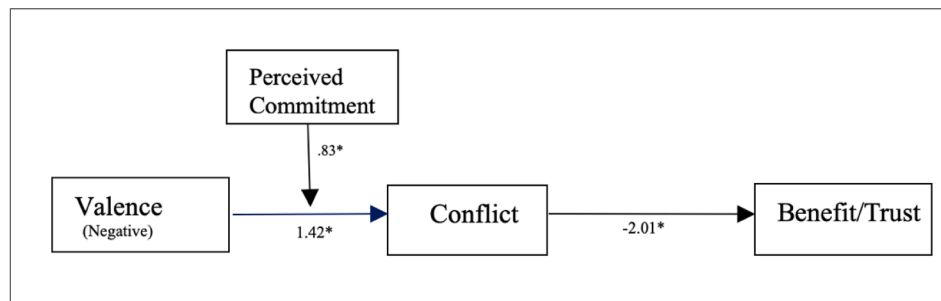


Fig. 2. Study 2–Mediation Figures Notes: * Coefficients significant at the 95% CI based on bootstrapping with 5000 samples.

0.17, $SE = 0.08$, $CI_{95}[-0.35, -0.04]$, and the interaction between review valence and perceived commitment was significant and positive ($b = 0.83$, $t = 4.28$, $p < 0.001$, see Table 4). The conditional indirect effect of negative valence on benefits and trust through perceived conflict was negative and significant when the perception of the airline's commitment was high but insignificant when the perception of the airline's commitment was low. Specifically, we conducted a floodlight analysis using values within the range of perceived commitment in the data. The Johnson–Neyman point had a perceived commitment of 3.34, meaning that the valence effect was insignificant ($p > 0.05$) when perceived commitment was smaller than 3.34. The indirect effect of negative (positive) reviews on benefits and trust was negative (positive) and significant when perceived commitment was > 3.34 . In these analyses, the effect of negative reviews on benefits and trust was negative and significant.

Study 2 demonstrates that although negative reviews do not affect benefits and trust directly, they increase readers' perceived conflict with a firm and therefore undercut benefits and trust. In addition, the results revealed that the detrimental effect is more potent when participants believed that a firm was committed to the relationship, giving marketers an indication of the importance of how to hold on to existing loyal consumers who are more vulnerable to negative information.

3.5. Study 3

3.5.1. Design and procedures

To test the effect of conflict on relational benefits and trust, we performed a lab experiment featuring the same experimental design used in the previous two studies but in a different service context: the hotel industry. We recruited another 190 Qualtrics members and gave them a nominal payment; 172 passed the attention checks ($M_{age} = 39$

years, 40% female).

To check the robustness of the effect of valence, we employed the same content approach for the stimuli, which is a different approach than the one we used in the two previous studies (Grewal & Stephen, 2019). All participants saw the same content under both negative and positive conditions. The only difference between the two conditions was the star ratings. The reviews had two stars in the negative and four in the positive conditions (see Appendix E). Since we mixed and balanced the negative and positive review texts, the same review could be treated as either positive or negative. To enhance realism, we took the language used in these reviews directly from TripAdvisor; half the reviews had a label displaying the number of reviews under each condition. Before they read the reviews, the participants were asked to indicate their emotional status using a 7-point scale.

Following the review task, we asked the participants to answer questions about perceived conflict (the same items from Study 1; $\alpha = 0.92$) and the relational benefits and trust of the hotel (the same items from Study 2; $\alpha = 0.87$). Then, in a random order, the participants answered attention checks and standard demographic questions.

3.5.2. Results

Relational benefits and trust. We predicted that the negative effect of conflict on relational benefits and trust would be more substantial for consumers with a higher level of perceived partner commitment. To test this, we regressed relational benefits and trust on conflict, perceived commitment, and their interaction. The overall model was significant ($F(3,168) = 5.76$, $p < 0.001$). There was both a main effect of conflict ($b = 0.71$, $t = 2.70$, $p = 0.008$) and perceived commitment ($b = 0.65$, $t = 2.62$, $p < 0.01$). In addition, the interaction effect of conflict and perceived commitment was significant ($b = -0.17$, $t = -3.43$, $p < 0.001$, see Table 5).

Table 4
Moderated Mediation Table, Study 2.

-		M (Conflict)		p-value	Y (Benefits and Trust)		
		Estimate			Estimate		p-value
Primary Predictors							
	X (Valence_Negative)	−2.18***	(1.02)	<0.001	−1.01	(0.69)	0.15
	M (Conflict)	−	−	−	−0.20**	(0.07)	0.006
	W (Perceived Commitment)	−1.05***	(0.15)	<0.001	0.001	(0.14)	0.99
	Valence (Negative) × P.Commitment	0.83***	(0.19)	<0.001	−	−	−
Controls							
	Intercept	7.65***	(1.02)	<0.001	5.56***	(0.96)	<0.001
	Age	−0.016	(0.01)	0.26	−0.02**	(0.01)	0.06
	Gender (Female)	0.24	(0.23)	0.30	−0.36**	(0.18)	0.05
	Education Level	0.03	(0.09)	0.40	0.01***	(0.07)	<0.001
	Emotional Status	0.07	(0.08)	0.41	0.27***	(0.05)	<0.001
Model Fit							
	R-square	0.686			0.564		
	F-statistic	14.75 (7,116)***	p-value < 0.001		6.70 (8,115)***	p-value < 0.001	

* $p < 0.10$. ** $p < 0.05$ *** $p < 0.001$.

Notes: Standard errors are in parentheses.

Table 5
Moderated Mediation Table, Study 3.

	M (Conflict)			Y (Benefits and Trust)		
	Estimate		p-value	Estimate		p-value
Primary Predictors						
X (Valence_Negative)	-1.01	(0.79)	0.17	-0.10	(0.16)	0.54
M (Conflict)	-	-	-	0.70**	(0.26)	0.008
W (Perceived Commitment)	-0.61***	(0.11)	<0.001	0.64**	(0.25)	0.01
Valence (Negative) × P.Commitment	0.35**	(0.16)	0.028	-	-	-
Conflict × P.Commitment	-	-	-	-0.17***	(0.05)	0.001
Controls						
Intercept	6.84***	(0.78)	. < 0.001	1.82	(1.47)	0.22
Age	-0.02**	(0.01)	0.008	-0.01	(0.01)	0.12
Gender (Female)	-0.33*	(0.18)	0.07	-0.09	(0.16)	0.57
Education Level	0.13*	(0.08)	0.098	-0.06	(0.07)	0.39
Emotional Status	0.09	(0.06)	0.17	0.25***	(0.05)	<0.001
Model Fit						
R-square	0.552			0.464		
F-statistic	10.27 (7,164)***	p-value < 0.001		5.60 (8,163)***	p-value < 0.001	

* $p < 0.10$. ** $p < 0.05$ *** $p < 0.001$.

Notes: Standard errors are in parentheses.

Moderated mediation analysis. We used the PROCESS Model 58 (Hayes, 2018) to further examine whether the interaction effect held in the entire moderated mediation model. We noted a significant interaction between review valence and perceived commitment ($b = 0.35$, $t = 2.22$, $p = 0.028$), as well as between conflict and perceived commitment ($b = -0.17$, $t = 4.69$, $p < 0.001$). While the direct effect of valence on benefits and trust was not significant ($b = -0.10$, $SE = 0.16$, $CI_{95}[-0.40, 0.21]$), the conditional indirect effect of negative valence on benefits and trust, through perceived conflict, was negative and significant when the perception of the hotel's commitment was high, but insignificant when the perception of the hotel's commitment was low.

We conducted a follow-up floodlight analysis using values of perceived commitment from the data. The Johnson–Neyman point had a perceived commitment of 4.24, meaning that the valence effect was insignificant ($p > 0.05$) when perceived commitment was smaller than 4.24. Conversely, the indirect effect of negative (positive) reviews on benefits and trust was negative (positive) and significant when perceived commitment was > 4.24 .

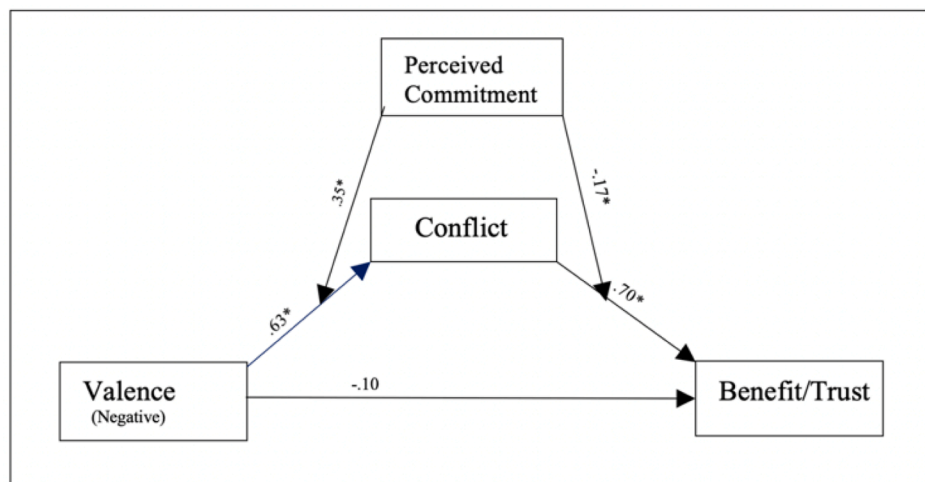
Study 3 indicates that although negative reviews do not affect benefits and trust directly, they increase readers' perceived conflict and therefore undercut benefits and trust (see Fig. 3). In addition, the results revealed that the detrimental effect was stronger when the participants

believed that the focal firm was committed to the relationship, giving marketers an indication of the importance of how to hold on to existing loyal consumers who are more vulnerable to negative information.

4. General discussion

4.1. Theoretical contributions

This paper's theoretical implications provide insight into the practicality of SET in online contexts. Our results from the three studies suggest that a balanced relationship involves a series of trade-offs between exchange partners. Adding a slight weight (e.g., a piece of negative information) on one side could deteriorate perceived trust and trigger a relational rebalancing process. The present work provides complementary evidence of how traditional relationship marketing theories could be applied to build online relationships. This reaffirms that eWOM is not only an outcome of marketing programs but is also an essential antecedent for building relational trust, especially for consumers with a higher level of commitment to the relationship. Given the rising influence of UGC and the crucial role of eWOM in the consumer decision-making process and RM, it is necessary to examine how consumers respond to different types of information when processing online



Notes: * Coefficients significant at the 95% CI based on bootstrapping with 5000 samples

Fig. 3. Study 3–Mediation Figures.

reviews. Despite extensive research on online reviews, how those reviews affect consumers' perceived relationships with firms is largely understudied (De Bruyn & Lilien, 2008). Across the three studies in different service categories, this work provides statistical evidence of a systematic linkage between review valence and consumers' perceived benefits and trust in a relationship. First, negative reviews (vs. positive ones) cause intensified conflict with the service provider. Second, because of the increased level of conflict, consumers feel the relationship is less beneficial, and the service provider is less truthful. Finally, a negative review is more persuasive for those who believe the service provider is firmly committed to the relationship. This adds another insight into prior literature on consumers' responses to negative information. For example, Ahluwalia et al. (2000) found that negative information for highly committed consumers with a brand is less diagnostic, and therefore they are less affected by it. However, our findings suggest that the diagnosis reducing does not occur when consumers believe the other side's commitment decreases.

These studies call for more research on how emerging new media affect online relationships (Hennig-Thurau et al., 2010; Steinhoff et al., 2019). The results from all three studies with different service contexts are consistent. These steady patterns, demonstrated across studies, provide statistical evidence that online reviews are antecedents of an existing online relationship. The results also suggest that although reviews do not directly affect consumers' perception of a relationship, the relationship is altered by an indirect path. These findings reconcile previous RM studies that looked at WOM solely as an outcome of RM, implying that improved benefits and trust will positively affect consumers' intention to engage in PWOM (Palmatier et al., 2006; Steinhoff et al., 2019). This work provides statistical evidence that the causal linkage between the B2C relationship and WOM does not stop when consumers post positive or negative WOM; instead, the new WOM will cause another wave of effects that will either enhance or undercut existing relationships with the firm for consumers who are exposed to the information. WOM serves as both an antecedent and a consequence of a relationship, and there is a circulated chain effect between RM and WOM, which researchers must consider when gauging the magnitude of WOM. Moreover, previous RM literature has focused on relationships manifested in the B2B context; this work provides additional insight into how UGC affects consumers' perceptions in the B2C context, which is critical for firms that pursue consumer lifetime value as a strategic goal.

4.2. Managerial implications

Our results provide additional evidence about how critical it is for managers to establish and maintain a reciprocal online relationship with consumers. Previous research has primarily focused on generating positive eWOM due to marketing activities (R. Liu et al., 2022). To supplement the data, we aimed to identify the valence of online information as a significant antecedent of online relationships; this signals to marketers that they should pay special attention to the indirect effect of online information valence and adjust their marketing strategies accordingly. For example, it would be justifiable to allocate an elevated marketing budget to strengthen capacity in tackling negative online information, given the underestimated assessment of the overall effect of valence that current studies have demonstrated.

In addition, the results indicate that negative online information especially hurts consumer groups with a high level of commitment. The moderating role that perceived commitment plays in the effect of review valence implies that consumers who have stronger beliefs in a firm are more vulnerable to negative information, highlighting the importance of consistent communication with existing consumers and using positive information to build a reciprocal relationship, thereby reaffirming firms' commitment to their customers. This is especially important because the marketing literature has revealed that recruiting new customers is five times more expensive than maintaining existing ones, who are often neglected by firms' marketing efforts. For example, many cable

companies make a compelling effort to recruit new customers by sharing information about their services' advantages and rewarding new customers with an attractive incentive package while neglecting those who have used their services for many years. This type of marketing practice might be beneficial in expanding a firm's customer base. However, in light of our findings, it is not necessarily helpful for the overall outcome of a firm's long-term marketing strategy. To avoid this inefficiency, we suggest that firms re-emphasize recognizing the wants and needs of their committed existing customer groups.

5. Conclusions

Although the study established an indirect path through review valence and relational benefits and trust and the moderating impact of perceived commitment on the process, we encountered limitations in scope and depth that indicate multiple avenues for future research. First, using eWOM valence as a single attribute for testing eWOM impact needed improvement. Other attributes of the eWOM matrix (e.g., volume, variance, length) are essential in determining the magnitude of eWOM's impact (Babić Rosario et al., 2016). Future research could examine the effect of review valence on relational constructs in the presence of additional elements. For example, an interesting research question would be whether a larger volume of negative reviews exacerbates a relationship proportionately. Second, the present research is rooted in online reviews from rating websites, which paves the way for future research to check our findings' robustness and external validity across different online platforms, such as social media and e-commerce websites. For example, as more and more firms employ social media as their primary marketing strategy, it would be interesting to investigate how social media UGC affects consumers' perceived relationship with a focal business. Third, culture plays a critical role in international RM; thus, reciprocal relationships may manifest differently in varied cultural environments (Samaha et al., 2014; Shaalan et al., 2013). As such, future studies should investigate the eWOM-RM interaction in cross-cultural contexts. Finally, we used a cross-subject research design with valence manipulation, in which we treated valence as a categorical variable, ignoring that it could be a continuous vector with mixed elements in reality (i.e., negative content in a positive review). In other words, scrutinizing valence only based on a dichotomous approach (positive or negative) could be insufficient regarding explanatory power. Future research should investigate the complexity of different aspects of valence and develop a new Likert scale grounded in multi-continuous items, thus reconfirming the robustness of the abovementioned results.

CRedit authorship contribution statement

Ran Liu: Software, Methodology, Formal analysis, Data curation, Conceptualization, Investigation, Project administration, Resources, Validation, Writing - original draft, Writing - review & editing. **John B. Ford:** Supervision, Conceptualization, Formal analysis, Methodology, Project administration, Writing - original draft, Writing - review & editing. **Weiyong Zhang:** Supervision, Formal analysis, Conceptualization, Data curation, Methodology, Writing - original draft. **Joseph Bonnici:** Supervision, Conceptualization, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendices.

Appendix A Review stimuli (Study 1)

	Positive Valence	Negative Valence
Manipulated review	 -This bank is great! They have great locations. They also offer state-of-the-art services that help you avoid standing in line.  I am typically leery of banks in general, but the bank uses advanced technology to protect my privacy, I do feel very secure with the bank.  The bank has an excellent mobile app. You can quickly transfer funds and set up to be notified of any account activity immediately as it occurs. At my local branch, the customer service agents are friendly and helpful.	 -This bank is terrible! They have poor locations. They also offer very old-fashioned services that make you stand in line for a long time.  I am typically not leery of banks in general, but the recent data breach report makes me feel unsafe with my account info.  The mobile app is terrible. It's so difficult to transfer funds, and there is no way to set up to be notified of any account activity immediately as it occurs. The customer service agents are rude and unhelpful.

Appendix B: Model 3

The regression model was as follows $Y_c = \alpha_0 + \alpha_1 V + \alpha_2 C_m + \alpha_3 VC_m + \alpha_i \sum \text{control variables} + \varepsilon$ (Model 3).

where Y_c is the estimation of consumers' conflict, the coefficient α_s is the OLS parameter estimate of the regression coefficients, V is the review valence, C_m is the perceived partner's commitment to the relationship, VC_m is the interaction of valence and commitment. Control variables such as age, gender, education level, and emotional status are included. ε is the error term. We chose these control variables because prior research indicates that in addition to demographic traits, emotional status may impact eWOM behavior (Burger, 2011; H. Liu et al., 2021).

Appendix C: Review stimuli (Study 2)

	Positive Valence	Negative Valence
Manipulated review	 Their customer service is fantastic. All agents are sweet and kind. Especially the gate agents, they are so patient and professional, and they treat passengers with respect.  Although my connection flight was delayed for 4 h, they provided free food and compensation. A customer agent came to me several times, asking me if I needed any help. I think I will fly again with this airline.  The flight was very clean, and the crew was professional! I found the meals were excellent. There were a variety of meals to choose from, wine and liquor were also provided. Overall, the service, entertainment, and professionalism made this a pleasant flight.	 Their customer service is abysmal. All agents are rude and threatening. Especially the gate agents, they are ignorant, and they couldn't even be bothered to answer my question!  They delayed my connection flight, and I had to wait in the airport for an additional 4 h. There was no compensation, and they only gave us some water. I think I won't use this airline if there is another option.  They have a dirty cabin and hungover crew! Do you like cleaning your seat of some fresh organic stain? Then this might be the airline for you as they gave me a hot towel to clean it myself! And the food tastes terrible, and you don't have a choice.

Appendix D: Model 2 (Study 2)

Effects of review valence on relational benefits and trust. To test H2, we used a moderated mediation model with 5,000 bootstrapping mediation methods, for which we conducted a randomized sample replacement process 5,000 times from the original sample pool with 155 participants (Preacher et al., 2007). In particular, we used the PROCESS Model 8 (Hayes, 2018), which can be explained mathematically below (Models 2a and 2b):

$$Y_c = \alpha_0 + \alpha_1 V + \alpha_2 C_m + \alpha_3 VC_m + \alpha_i \sum \text{control variables} + \varepsilon \text{ (Model 2a).}$$

$Y_{t,b} = \beta_0 + \beta_1 V + \beta_2 C_m + \beta_3 VC_m + \beta_4 Y_c + \beta_i \sum \text{controlvariables} + E$ (Model 2b). where $Y_{t,b}$ is the estimation of consumers' benefits and trust, the coefficient β s are OLS parameter estimates of the regression coefficients, V is the review valence, C_m is the consumer's level of commitment to the relationship, VC_m is the interaction of valence and commitment, and the variables within parentheses are those stated in Model 1 of Study 1. Control variables such as age, gender, education level, and emotional status are included. ϵ and E are the error terms.

In Model 3b, the effect of conflict on benefits and trust is β_4 ; the direct effect of valence benefits and trust is β_1 , and the indirect effect is $\beta_4\alpha_1$. Both direct and indirect effects are expected to be negative (when using a positive review group as a reference group); however, their magnitude is determined by the level of commitment. Spotlight analyses are expected to show that the magnitude of the effect of valence (β_1) will be weaker when the level of commitment is low.

Appendix E. Review stimuli (Study 3)

	Positive Valence	Negative Valence
Manipulated review	The hotel has clean rooms and friendly staff. You have plenty to do for both adults and kids. There were pool games, big gyms, water features, and breakfast was great. However, the hotel had some issues in management; it took us about thirty minutes to check in because the front desk staff had to ask housekeepers to find out which room was available.	The hotel has clean rooms and friendly staff. You have plenty to do for both adults and kids. There were pool games, big gyms, water features, and breakfast was great. But the hotel had some issues in management; it took us about thirty minutes to check in because the front desk staff had to ask housekeepers to find out which room was available.

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Dr. Ran Liu is an Assistant Professor in the Marketing department at Central Connecticut State University (CCSU). He holds a Ph.D. in Marketing with a minor in International Business from Old Dominion University, and his research is centered around online relationships, user-generated content (UGC), and international business. Dr. Liu also serves as the Associate Editor (Asia) for the Journal of Eastern European and Central Asian Research (JEECAR) and as a Faculty Senator at CCSU.

Dr. John Ford is a professor of Marketing and International Business, Eminent Scholar, and Haislip-Rorher Fellow at Old Dominion University. He earned his doctoral degree from the University of Georgia. His research interests are in the areas of international advertising strategy, consumer reaction to advertising exposure, cross-cultural marketing research issues, and non-profit donor behavior. He is presently serving as the Editor-in-Chief for the Journal of Advertising Research and is also an Associate Editor for Advertising Research for the Journal of Business Research. He serves on the Editorial Review Boards for the Journal of the Academy of Marketing Science, Journal of Advertising, International Marketing Review, Journal of International Marketing, International Journal of Advertising, Marketing Education Review, International Journal of Nonprofit and Voluntary Sector Marketing, Journal of Marketing Theory and Practice, Cornell Hospitality Quarterly and Asia Pacific Journal of Marketing and Logistics. He is a Past-President of the Academy of Marketing Science and serves on the Board of Governors.

Dr. Weiyong Zhang is an Associate Professor in the Department of Information Technology and Decision Sciences at Strome College of Business, Old Dominion University. He holds both a Bachelor's and Master's degree in Management Information Systems from Fudan University, China, and a Ph.D. in Operations and Management Sciences from the University of Minnesota. He accumulated extensive industrial experience while he worked as a consultant at Hewlett-Packard. His research interests include information systems, project management, process improvement, supply chain management, and quantitative methods. His work has appeared in top-tier research journals such as the International Journal of Information Management, Internet Research, Information Systems Frontier, Journal of Global Information Management, Electronic Markets, Technological Forecasting and Social Change, International Journal of Forecasting, Quality Management Journal, and Project Management Journal.

Dr. Joseph Bonnici holds a Ph.D. in Business Administration from the University of Georgia, a JD from Roger Williams University's Law School, and an MBA from Georgia Southern University. He joined CCSU as a full professor in 2002 and was initially active in the CCSU-Siracusa Academic Consortium program in Europe. Since then, he has organized several study abroad programs for our students. With the cooperation of other CCSU faculty members, he transformed the International Journal of Arts and Sciences (IJAS) into the world's largest multidisciplinary conference platform in academia. He serves on several editorial boards. He is a federally licensed attorney with membership in different bars including Connecticut's and the District of Columbia's. He has also served as a solicitor in England and Wales.